

B.Tech. Project Mid Sem Presentation on

Satellite-Based Cyclone Detection and Intensity Estimation in the Indian Ocean Region

Presented by

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Project Guide

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Why Cyclone Detection?

- Cyclones are one of the most devastating natural disasters affecting India, particularly along its eastern and western coastlines.
- As many as 117 cyclones hit India in 50 years from 1970-2019 claiming over 40,000+ lives

Why Cyclone Detection advancement is required?

- Traditional methods: The Dvorak method
- Manual interpretation of satellite images.
- Time-consuming, Subjective, and limited in analysis.

Traditional Methods for Cyclone Detection

- Dvorak method
- Advanced Dvorak Technique (ADT)
- Deviation-Angle Variance Technique (DAVT), etc...

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Early studies pioneered the use of satellite data for cyclone genesis detection and center localization

Subsequent advancements

- Introduced CNNs for intensity estimation.
- Hybrid models for multi-sensor fusion.
- Recent innovations leverage cutting-edge architectures like Faster R-CNN and YOLOv8 for real-time tracking.
- Integration of artificial intelligence (AI) & Deep learning (DL): Enabling automated, real-time, and high-accuracy frameworks for cyclone detection.

Overview of Proposed Work

- Satellite Image Processing : GOES series or INSAT-3D
- Web Scrapping :
 - Automatic retrieve GOES satellite imagery and storm outlook data from NOAA.
 - Real-time and historical cyclone data for analysis.
- Deep Learning Frameworks: TensorFlow or PyTorch (used to design, train CNN Model)
- Flask API : Data requests, Model inference, and Response generation.
- React Dashboard : For front-end user interface (UI)

Traditional Methods & Early Satellite Data Utilization (Pre-2010s)

- N Uma et al., Dvorak Technique, Advanced Dvorak Technique (ADT), and Passive Microwave Imagery: Manual analysis of cloud patterns and thermal infrared (IR) data.
- Piñeros et al., Kalman Filters and statistical approaches like axisymmetric cloud organization analysis were used for cyclone tracking and genesis prediction, achieving 93% true positive rates but with high false alarms.
- Talukder Ashit et al., QuikSCAT (wind data) and TRMM (precipitation data) enabled global cyclone detection, achieving a 91.67% true positive rate.

Machine Learning Integration (2010s)

- MD. Haque Nazmul et al., Used support vector machines (SVM), decision trees (DT), and random forest (RF) achieved 86–95% accuracy in cyclone classification.
- The ISCCP and NOAA GridSat dataset standardized geostationary satellite data (1980–2011) for climate studies, enabling global cyclone tracking.
- Shakya et al., Linear regression and optical flow techniques improved cyclone path forecasting and temporal resolution.

Early Deep Learning Adoption (Late 2010s–Early 2020s)

- Kavitha D., CNNs emerged for spatial feature extraction, achieving 95% detection accuracy using IR/WV satellite imagery
- Suresh Anjana, Hybrid CNN-RNN models combined spatial and temporal data, reducing RMSE to 3.58 knots
- YANG Wen et al., Pre-trained CNNs (VGGNet, ResNet) achieved a 4.05 m/s MAE in intensity estimation, outperforming the Dvorak technique
- Xie et al., YOLOv4 and Faster R-CNN were applied for cyclone eye localization, achieving 87.86% mAP

Advance Deep Learning Adoption (2020s - 2024s)

- Shakya et al., RetinaNet and LSTM hybrid models achieved 97% classification accuracy with optical flow interpolation.
- Hybrid CNN + YOLOv3 models achieved 98.4% binary classification accuracy for North Indian Ocean cyclones. [An End-to-End Deep Learning Framework, 2024]
- Xie et al., Data fusion of ASCAT wind fields and precipitation radar data enhanced global detection accuracy using FPN + Faster R-CNN.
- Chang-Jiang Zhang et al., Global hybrid models using GOES-16 satellite imagery achieved 94% accuracy 12 hours before landfall

Research Gap Analysis

- 1 Many existing studies have advanced models like ViTs achieve high accuracy but require significant computational resources, limiting real-time deployment.
- 2 High-resolution satellite systems are costly and inaccessible for developing regions.
- 3 Places like the North Indian Ocean have fewer cyclone records, making AI models less accurate there.
- 4 Early cyclone detection systems often mistake normal storms for cyclones (false alarms).

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Traditional methods for cyclone detection often rely on manual analysis and heuristic approaches, which can be slow and inaccurate. Our study shows for more automated and efficient cyclone detection systems that take advantage of the vast amounts of satellite imagery and meteorological data available. using deep learning approach like CNN for better cyclone intensity estimation and RNN and LSTM for path prediction.

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- ➊ To make a survey of existing techniques that are proven sound enough to cyclone detection
- ➋ Develop a Deep Learning-Based Cyclone Detection System.
- ➌ Improve the prediction of the intensity and path of the cyclone using new approaches like CNN and LSTM
- ➍ Implement a Real-Time Cyclone Detection using real-time satellite images

① Data Collection & Preprocessing

- Data Retrieval
 - web scraping to collect real-time satellite imagery
 - Image Data - GOES or INSAT 3D satellite
- Preprocessing:
 - Normalize, enhance, and clean the satellite imagery.

② Model Development

- convolutional neural network (CNN)

③ Training & Validation:

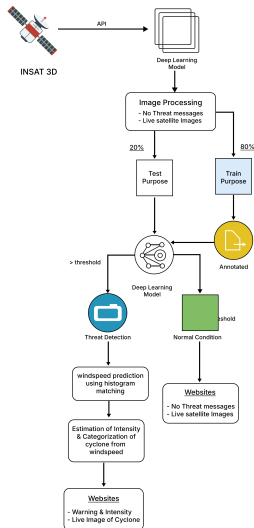
- CNN on historical cyclone data on 80% data.
- Testing: on 20% of original data.

④ Backend Development (Flask API):

- a RESTful API using Flask to handle data requests, process satellite imagery, and return cyclone intensity predictions via defined endpoints.

⑤ Frontend Implementation (React Dashboard)

Proposed Methodology Contd...



Cyclone Intensity Prediction

Result and Discussion

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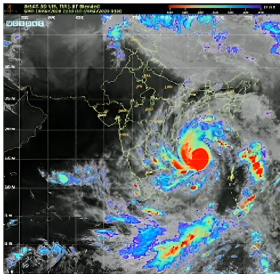
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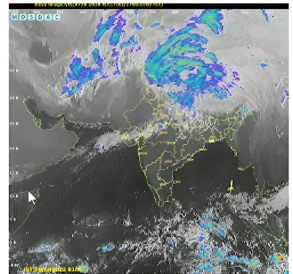
References

The live images captured by INSAT-3D Satellite were retrieved.

Cyclone



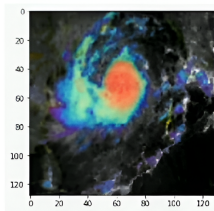
No Cyclone



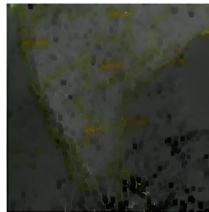
Result and Discussion Contd...

The important features of the image were extracted using dilation and erosion before passing it into the model. This step is necessary so as to improve the model performance.

Cyclone



No Cyclone



Result and Discussion Contd...

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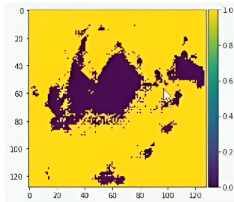
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After extracting the important features the images were passed through CNN model which distinguished the low pressure region by turning it into violet colour and the background into yellow.

Cyclone



No Cyclone



Result and Discussion Contd...

The Wind speed was found with correlation of histogram data between test and trained image is 119 knots.

Serial No.	T number	Classification of Cyclonic Disturbance	Wind Speed	Wind Criteria	Pressure depth
1	T1.0	Low	-	<17	-
2	T1.5	Depression	25	17-27	-
3	T2.0	Deep Depression	30	28-33	4.5
4	T2.5	Cyclonic Storm	35	34-47	6.1
5	T6.0	Extremely Severe Cyclonic Storm	115	92-119	65.6
6	T6.5	Super Cyclonic Storm	127	> 120	80

Conclusion

- The study demonstrates the effectiveness of Convolutional Neural Networks (CNNs) in estimating tropical cyclone intensity using satellite infrared imagery
- The CNN-based system provides accurate and timely cyclone intensity predictions, enabling governments and communities to enhance disaster preparedness and response.
- Integration of CNN models with a React dashboard and data storage advances extreme weather impact management.
- Challenges remain in refining prediction methods, addressing time constraints, and expanding regional model generalization..

Future Scope

- Embedded systems, powered by AI/ML, advanced modeling, and multi-source data, will deliver more accurate, timely cyclone predictions to improve disaster preparedness.
- Cross-sector collaboration and technological advancements (e.g., high-performance computing) will drive innovation in cyclone forecasting tools and methodologies.

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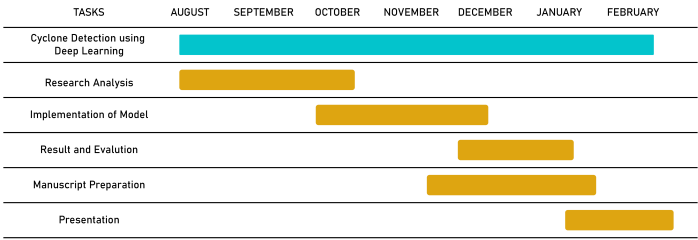


Figure 1: Timeline Chart

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- 1 Uma N, Jithin Vinod, K M Komal, Joshua James, et al. "Deep Learning Based Cyclone Intensity Estimation using INSAT-3D IR Imagery" (2024).
- 2 Anjana Suresh. "CYCLONE INTENSITY ESTIMATION USING INSAT 3D IR IMAGERY BASED ON DEEP LEARNING"
- 3 Harshal Namdeorao Dharpure, Tejal Sudhakar Rao Mohod, Radhika Vinod Malani, Janhavi Chandak, Atharva Shekhar Belge, Preet Ravin Ambadkar. "Deep Learning-Based Cyclone Intensity Estimation Using INSAT-3D IR Imagery" A Comparative Study
- 4 Md.Nazmul Haque, Ashfaquul Adel, Kazi Saeed Alam. "Deep Learning Techniques in Cyclone Detection with Cyclone Eye Localization Based on Satellite Images" (2021)
- 5 Pradeep K. Goyal. "Cyclone Disaster Mitigation and Management in India"
- 6 M. SWARNA, N. SUDHAKAR, N. VADAPARTHI. "An effective tropical cyclone intensity estimation model using Convolutional Neural Networks".
- 7 Atharv Dabhade; Samshitha Roy; Marwa S. Moustafa; Sayed A. Mohamed; Reda El Gendy; Shovan Barma, et al. "Extreme Weather Event (Cyclone) Detection in India Using Advanced Deep Learning Techniques"
- 8 Akhila G Nair. "A study on various tropical cyclone hits in India – through gis approach"
- 9 Snehlata Shakya , Sanjeev Kumar, Mayank Goswami. "Deep Learning Algorithm for Satellite Imaging Based Cyclone Detection"
- 10 Ming Xie , Ying Li, Shuang Dong, et al. "A Deep-Learning-Based Fusion Approach for Global Cyclone Detection Using Multiple Remote Sensing Data" (2022).
- 11 Miguel F. Piñeros, Elizabeth A. Ritchie, J. Scott Tyo, et al. "Detecting Tropical Cyclone Genesis From Remotely Sensed Infrared Image Data"

References Contd...

- 12 Kavitha D, Nikhil T, Praveen R, Mohammed Nhj. "Detection of Tropical Cyclone using Deep Learning Network"
- 13 Marwa S. Moustafa, Mohamed R. Metwalli, Roy Samshitha, Sayed A. Mohamed, Barma Shovan. "Cyclone detection with end-to-end super resolution and faster R-CNN"
- 14 Austin Harris, Shijia Wei, Prateek Sahu, Pranav Kumar, Todd Austin, Mohit Tiwari. "Cyclone Detecting Contention-Based Cache Information Leaks Through Cyclic Interference"
- 15 Buo-Fu Chen, Boyo Chen, Hsuan-Tien Lin, and Russell L. Elsberry. "Estimating Tropical Cyclone Intensity by Satellite Imagery Utilizing Convolutional Neural Networks"
- 16 Chitti Anusha, Kothuru Sai Sanjay, Shenkeshi Deekshitha, Saroja Kumar Rout, Kottu Santosh Kumar, S Ranjith Reddy, et al. "Super Cyclone Detection and Tracking Using YOLOv8 Algorithm"
- 17 Swapandeep Kaur et al. "Hurricane Damage Detection using Machine Learning and Deep Learning Techniques: A Review"
- 18 Kenneth R. Knapp, Steve Ansari, Caroline L. Bain, Mark A. Bourassa, Michael J. Dickinson, Chris Funk, Chip N. Helms, Christopher C. Hennon, Christopher D. Holmes, George J. Huffman, James P. Kossin, Hai-Tien Lee, Alexander Loew, Gudrun Magnusdottir. "Globally Gridded Satellite Observations for Climate Studies"
- 19 Manish Kumar Mawatwal, Saurabh Das. "An End-to-End Deep Learning Framework for Cyclone Intensity Estimation in North Indian Ocean Region Using Satellite Imagery"
- 20 Anand Panangadan, Shen-Shyang Ho, Ashit Talukder. "Cyclone Tracking using Multiple Satellite Image Sources"

References Contd...

- 21 Manil Maskey, Rahul Ramachandran, Muthukumaran Ramasubramanian, Iksha Gurung, Brian Freitag, Aaron Kaulfus, et al. "Deepti: Deep-Learning-Based Tropical Cyclone Intensity Estimation System"
- 22 Manil Maskey, Rahul Ramachandran, Iksha Gurung, Brian Freitag, J. J. Miller, Muthukumaran Ramasubramanian. "Machine Learning Lifecycle for Earth Science Application: A Practical Insight"
- 23 Shen-Shyang Ho, Ashit Talukder. "Automated Cyclone Discovery and Tracking using Knowledge Sharing in Multiple Heterogeneous Satellite Data"
- 24 Chitti Anusha, Kothuru Sai Sanjay, Shenkeshi Deekshitha, Saroja Kumar Rout, Kottu Santosh Kumar, S Ranjith Reddy. "Super Cyclone Detection and Tracking Using YOLOv8 Algorithm"
- 25 Neeru Jaiswal, C. M. Kishtawal, et al. "Objective Detection of Center of Tropical Cyclone in Remotely Sensed Infrared Images"
- 26 Jia Liu, Chuancai Liu, Boyang Wang, Danyu Qin. "A novel algorithm for detecting center of tropical cyclone in satellite infrared images"
- 27 Zhen Wang, Jun Zhao, Hong Huang, Xuezhong Wang. "A Review on the Application of Machine Learning Methods in Tropical Cyclone Forecasting"
- 28 Shweta Kumawat, Jitendra Jaiswal. "Cyclone Detection and Forecasting Using Deep Neural Networks Through Satellite Data"
- 29 Sudha Radhika, Yukio Tamura, Masahiro Matsui. "Cyclone damage detection on building structures from pre- and post-satellite images using wavelet based pattern recognition"

References Contd...

- 30 Ashit Talukder, Shen-Shyang Ho, Timothy Liu, Wendy Tang, Andrew Bingham, Eric Rigor. "Global Cyclone Detection and Tracking using Multiple Remote Satellite Data"
- 31 Kavitha D, Nikhil T, Praveen R, Mohammed Nhj. "Detection of Tropical Cyclone Using Deep Learning Network"
- 32 Veena R S, Abhijna K C, B G Shreyas, Bhargavi, Dhanush Gowda S. "Deep learning and INSAT-3D IR Imagery for Estimating Cyclone Intensity"
- 33 Wen Yang, Jianfang Fei, Xiaogang Huang, Juli Ding, Xiaoping Cheng. "Enhancing Tropical Cyclone Intensity Estimation from Satellite Imagery through Deep Learning Technique"
- 34 Ritesh Pradhan, Ramazan S. Aygun, Manil Maskey, Rahul Ramachandran, Daniel J. Cecil, et al. "Tropical Cyclone Intensity Estimation Using a Deep Convolutional Neural Network"
- 35 Chang-Jiang Zhang, Xiao-Jie Wang, Lei-Ming Ma, Xiao-Qin Lu. "Tropical Cyclone Intensity Classification and Estimation Using Infrared Satellite Images With Deep Learning"
- 36 Johnson Sandra, B. Abdul Naeem, V. Benedict Vinusha, P. C. Charankumar, V. Indhuja. "CNN-Based Cyclone Intensity Estimation Using Satellite Imagery"

Questions